
Discrete Information Source and Self-Information

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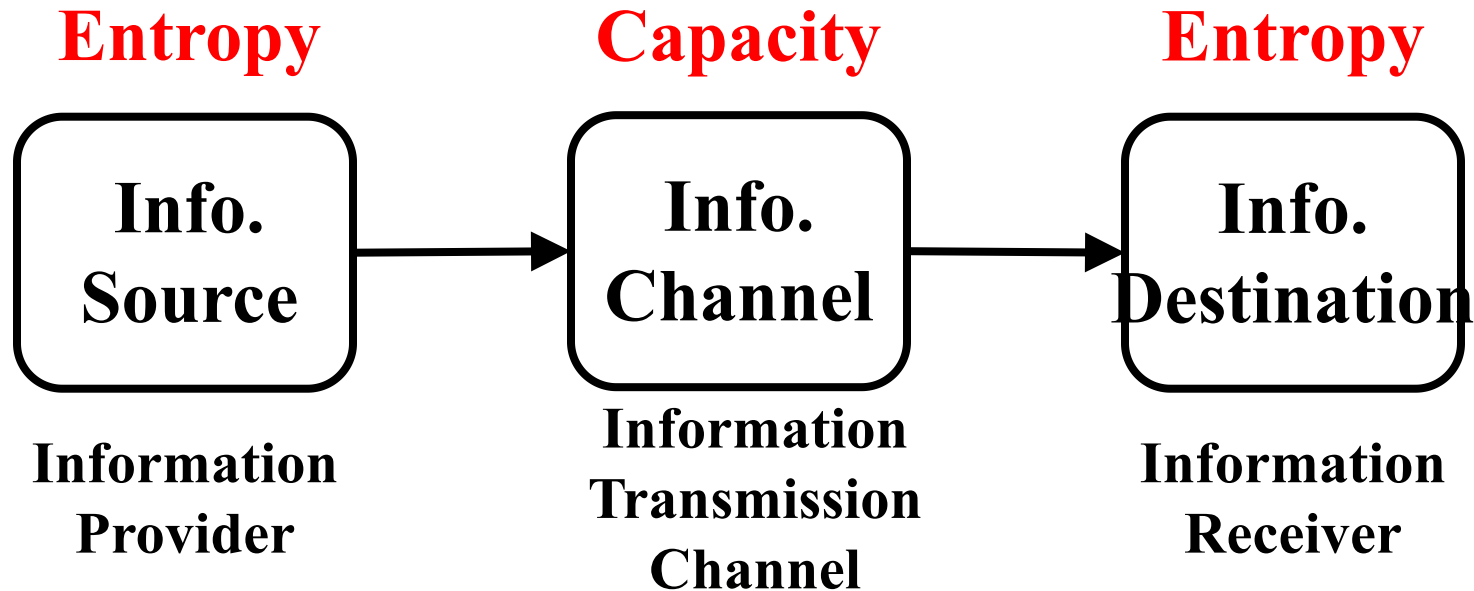
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级别		说明
一秀（秀才）	     	六个骰子中有一个点数是“4”
二举（举人）	     	六个骰子中有两个点数是“4”
四进（进士）	     	六个骰子中有四个点数是相同的
三红（探花）	     	六个色子里面有三个点数是“4”
对堂（榜眼）	     	六个骰子的点数刚好是顺子
状元	     	均为状元

Simplified Communication System



- ❑ Discussing the properties of information source.
- ❑ Studying how to model a specific information source and messages.
- ❑ Discussing how to quantify the information and its properties.

Main Content

- 1 Discrete Information Source**
- 2 Self-information of messages**
- 3 Properties of self-information**
- 4 Joint & Conditional self-information**

Discrete Information Source

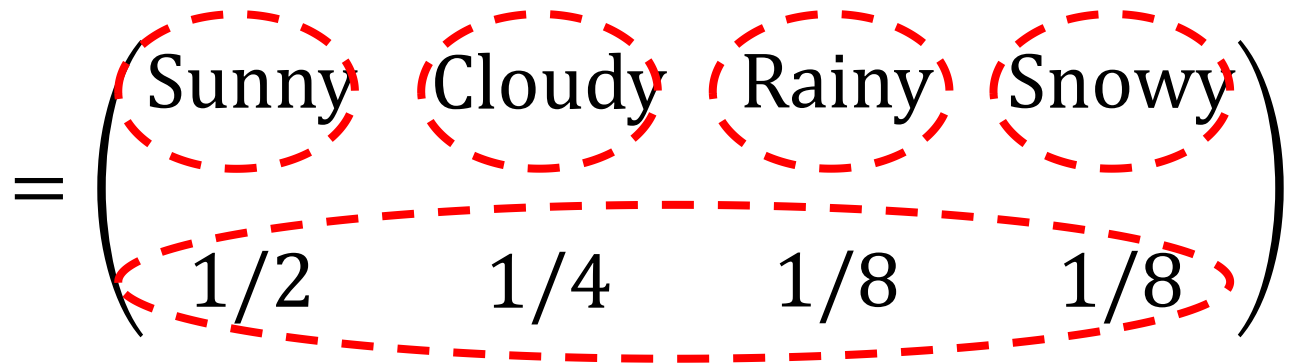
- ❑ Information source sends messages. Information is carried by uncertain/random messages. An information source is modelled by a **random variable/vector**, while messages sent by the information source are **legitimate values** taken by this random variable/vector.
- ❑ Discrete information source only sends **discrete messages**. This source has **finite choice** of sending messages. As a result, the message sequence sent by the discrete information source can be regarded as a random sequence having finite states.
- ❑ Examples: weather (sunny, cloudy, rainy, snowy), football game (win, draw, lose).
- ❑ If a discrete information source only concerns a single random event, it can be regarded as a **single-symbol** source; if a discrete source concerns multiple random events, it can be regarded as a **multi-symbol** source.

Example: Weather

**Discrete
Information
Source**



**Random
Messages**



**Statistical
Properties**

**Sending
Probabilities**

Math. Model of Discrete Info. Source

$$\begin{pmatrix} \text{Discrete Info. Source } X \\ \text{Prob. Space } P(X) \end{pmatrix} = \begin{pmatrix} \text{Msg } a_1 & \cdots & \text{Msg } a_i & \cdots & \text{Msg } a_n \\ \text{Prob. } p(a_1) & \cdots & \text{Prob. } p(a_i) & \cdots & \text{Prob. } p(a_n) \end{pmatrix}$$

❑ Info. Source sends messages, messages carry information, messages are random and uncertain.

❑ Probability Constraints:

$$0 \leq p(a_i) \leq 1, \quad p(a_1) + \cdots + p(a_i) + \cdots + p(a_n) = 1$$

Discrete Random Variables

- Probability Spaces of random variables X and Y :

$$\left(\begin{matrix} X \\ P(X) \end{matrix} \right) = \left(\begin{matrix} a_1, & \cdots & a_i & \cdots & a_n \\ p(a_1), \cdots p(a_i) \cdots p(a_n) \end{matrix} \right),$$

$$\left(\begin{matrix} Y \\ P(Y) \end{matrix} \right) = \left(\begin{matrix} b_1, & \cdots & b_i & \cdots & b_m \\ p(b_1), \cdots p(b_i) \cdots p(b_m) \end{matrix} \right)$$

- Joint random variable $XY \in \{a_i b_j | i = 1, 2, \dots, n; j = 1, 2, \dots, m; \}$, its probability is expressed as $p(a_i b_j)$. When X and Y are independent, their joint probability is $p(a_i b_j) = p(a_i) p(b_j)$.
- Conditional probability characterizes the inherent relationship between a pair of random variables. Given $X = a_i$, the probability of $Y = b_j$ is $p(b_j | a_i)$
- Probabilities, conditional probabilities and joint probabilities should satisfy a range of constraints and properties.

Regular news & Breaking news



Information Measurement—Regular news

Discrete Info.

2018-2019 ECL Semi 1st Round:

Source X :

Barcelona (0) vs Liverpool (0)

Message x_1 : Barcelona goes to the final, Odds: 1.65/1
Probability: $p(x_1)=58.5\%$

Regular News, Moderate Information.

Message x_2 : Liverpool goes to the final, Odds: 2.33/1
Probability: $p(x_2)=41.5\%$

Regular News, Moderate Information.

Information Measurement—Breaking news

Discrete Info.

2018-2019 ECL Semi 2nd Round:

Source Y :

Liverpool (0) vs Barcelona (3)

Message y_1 : Barcelona goes to the final, Odds: 1/50
Probability: $p(y_1)=99.9\%$

Expected, Regular News, Low Information.

Message y_2 : Liverpool goes to the final, Odds: 18/1
Probability: $p(y_2)=0.1\%$

Unexpected, Breaking News, Very High Information!

Information Measurement

One Coin



head



tail

Total States: $2 = 2^1$

Information: $1 = \log_2 2$
 $= -\log_2 \frac{1}{2}$

$1/2$ and $1/4$ are the occurrence probabilities of a specific state. Here, we assume all the states are equally distributed.

Two Coins



head



head



tail



head



head



tail



tail



tail

Total States: $4 = 2^2$

Information: $2 = \log_2 4$
 $= -\log_2 \frac{1}{4}$

Information Measurement

$$\begin{pmatrix} X \\ P(X) \\ I(X) \end{pmatrix} = \begin{pmatrix} a_1 & \dots & a_i & \dots & a_n \\ p(a_1) & \dots & p(a_i) & \dots & p(a_n) \\ I(a_1) & \dots & I(a_i) & \dots & I(a_n) \end{pmatrix}$$

Set of self-
information

Self-information of
every message

Self-Information is
quantified as:

$$I(a_i) = -\log_2 p(a_i) \text{ bit}$$

Information Unit

Self-Information is $I(a_i) = -\log_2 p(a_i)$ *bit*

□ Information unit is related to the base of logarithm:

- When the base is 2, information unit is '*bit*';
- When the base is *e*, information unit is '*nat*';
- When the base is 10, information unit is '*det*' or '*hart*'.

□ Equivalent transformation between information units:

$$1 \text{ Nat} = \log_2 e \approx 1.433 \text{ bit}$$

$$1 \text{ Hart} = \log_2 10 \approx 3.322 \text{ bit}$$

$$1 \text{ bit} = \ln 2 \approx 0.693 \text{ Nat}$$

$$1 \text{ bit} = \lg 2 \approx 0.301 \text{ Hart}$$

□ Information is a pure number without physical measurement unit.

Information Measurement—Breaking news

Discrete Info.

2018-2019 ECL Semi 1st Round:

Source X :

Barcelona (0) vs Liverpool (0)

Message x_1 : Barcelona goes to the final, Odds: 1.65/1
Probability: $p(x_1)=58.5\%$

Self-Information: $I(x_1) = -\log_2 58.5\% = 0.773 \text{ bit}$

Message x_2 : Liverpool goes to the final, Odds: 2.33/1
Probability: $p(x_2)=41.5\%$

Self-Information: $I(x_2) = -\log_2 41.5\% = 1.269 \text{ bit}$

Information Measurement—Breaking news

Discrete Info. **2019 UEFA Champions League Semi 2nd Round:**
Source Y : **Liverpool (0) vs Barcelona (3)**

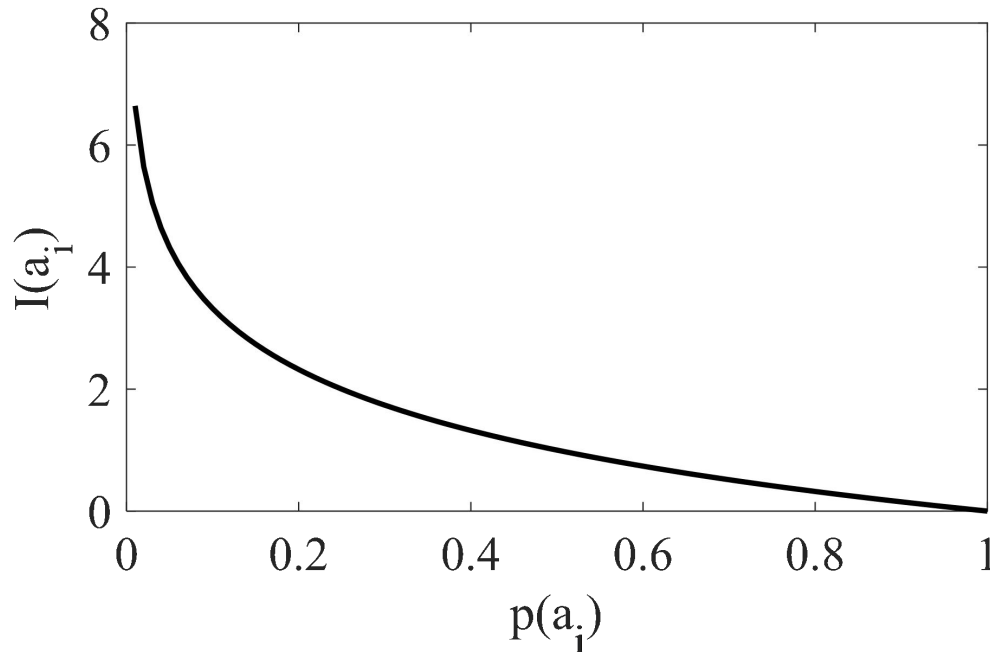
Message y_1 : Barcelona goes to the final, Odds: 1/50
Probability: $p(y_1)=99.9\%$

Self-Information: $I(y_1) = -\log_2 99.9\% = 0.0014 \text{ bit}$

Message y_2 : Liverpool goes to the final, Odds: 18/1
Probability: $p(y_2)=0.1\%$

Self-Information: $I(y_2) = -\log_2 0.1\% = 9.9658 \text{ bit}$

Properties of Self-Information



- Non-negative
- Monotonously decreasing
- If $p(a_i) = 1$, $I(a_i) = 0$
- If $p(a_i) = 0$, $I(a_i) = \infty$

❑ If an message has a high occurrence probability, it has low uncertainty and it thus has low information.

❑ The self-information $I(a_i)$ is a function of message a_i , it quantifies how much information conveyed by this message.

Example

“Double Color Ball” (双色球, *Shuangseiqiu*) is a popular lottery game in China. Each draw produces 7 numbers:

- **Red balls:** choose 6 distinct numbers from 1 to 33 (no repetition).
- **Blue ball:** choose 1 number from 1 to 16.

1. How difficult is it to correctly guess that the **first drawn ball** is number “7”?
2. After the **first ball** has been drawn, how difficult is it to correctly guess that the **second drawn ball** is number “16”?
3. Given that the first five drawn balls are “7, 16, 21, 17, 1”, how difficult is it to correctly guess that the **sixth drawn ball** is number “19”?
4. How difficult is it to correctly guess the complete Double Color Ball result:
Red: 7, 16, 21, 17, 1, 19 Blue: 14?



Joint Self-Information

□ Discrete information source concerning two random events:

$$\begin{pmatrix} XY \\ P(XY) \end{pmatrix} = \begin{pmatrix} a_1b_1, \dots a_1b_m, a_2b_1, \dots a_2b_m, \dots a_nb_1, \dots a_nb_m \\ p(a_1b_1), \dots p(a_1b_m), p(a_2b_1), \dots p(a_2b_m), \dots p(a_nb_1), \dots p(a_nb_m) \end{pmatrix}$$

where $0 \leq p(a_ib_j) \leq 1$, $\sum_{i=1}^n \sum_{j=1}^m p(a_ib_j) = 1$

□ The joint self-information of the joint message a_ib_j is:

$$I(a_ib_j) = -\log_2 p(a_ib_j)$$

□ When X and Y are independent with each other, the joint self-information of a_ib_j is:

$$I(a_ib_j) = -\log_2 p(a_i) - \log_2 p(b_j) = I(a_i) + I(b_j)$$

Conditional Self-Information

□ Conditional self-information is defined as the negative logarithm of the conditional probability.

□ Given b_j , the conditional self-information of a_i occurring is:

$$I(a_i/b_j) = -\log_2 p(a_i/b_j)$$

□ Given a_i , the conditional self-information of b_j occurring is:

$$I(b_j/a_i) = -\log_2 p(b_j/a_i)$$

□ Self-information, conditional self-information and joint self-information satisfy the following equalities:

$$\begin{aligned} I(a_i b_j) &= -\log_2 p(a_i) p(b_j/a_i) = I(a_i) + I(b_j/a_i) \\ &= -\log_2 p(b_j) p(a_i/b_j) = I(b_j) + I(a_i/b_j) \end{aligned}$$

Example: Self-Information Calculation

- ❑ In a community, 25% of the girls are undergraduate students. 75% of the female undergraduate students are 160 cm tall. Moreover, 50% of the girls living in this community is taller than 160 cm. If we receive a message ``a girl taller than 160cm is an undergraduate student'', how much information can we get from this message?
- ❑ **Solution:** Firstly, we need to derive the probability of the message ``a girl taller than 160cm is an undergraduate student''. Then, we may calculate its self-information with the aid of its definition.

Information Measurement—Breaking news

Discrete Info.

Source X : 2018 World Cup Group H: Germany vs Korea

Message x_1 : Germany wins Probability: $p(x_1)=81.3\%$

Expected, Regular News, Low Information.

Message x_2 : Draw Probability: $p(x_2)=13\%$

Unexpected, Attractive News, High Information.

Message x_3 : Korea wins Probability: $p(x_3)=5.7\%$

Unexpected, Breaking News, Very High Information!

Example – Breaking News

Discrete Info.

Source X : 2018 World Cup Group H: Germany vs Korea

Message x_1 : Germany wins Probability: $p(x_1)=81.3\%$

Self-Information: $I(x_1) = -\log_2 81.3\% = 0.3 \text{ bit}$

Message x_2 : Draw Probability: $p(x_2)=13\%$

Self-Information: $I(x_2) = -\log_2 13\% = 3 \text{ bit}$

Message x_3 : Korea wins Probability: $p(x_3)=5.7\%$

Self-Information: $I(x_3) = -\log_2 5.7\% = 4.1 \text{ bit}$

Example – Breaking News

Discrete Info.

2018 World Cup Final (90 mins):

Source Y :

France vs Croatia

Message y_1 : France wins Probability: $p(x_1)=50\%$

Self-Information: $I(y_1) = -\log_2 50\% = 1 \text{ bit}$

Message y_2 : Draw Probability: $p(x_2)=29\%$

Self-Information: $I(y_2) = -\log_2 29\% = 1.8 \text{ bit}$

Message y_3 : Croatia wins Probability: $p(x_3)=21\%$

Self-Information: $I(y_3) = -\log_2 21\% = 2.2 \text{ bit}$